TUTORIAL XIII

FOOTNOTES, MARGINPARS, AND ENDNOTES

LaTeX has facilities to typeset "inserted" text, such as footnotes, marginal notes, figures and tables. This chapter looks more closely at different kinds of notes.

XIII.1. FOOTNOTES

Footnotes are generated with the command

`\footnote{`*footnote_text*`}`

which comes immediately after the word requiring an explanation in a footnote. The text footnote_text appears as a footnote in a smaller typeface at the bottom of the page. The first line of the footnote is indented and is given the same footnote marker as that inserted in the main text. The first footnote on a page is separated from the rest of the page text by means of a short horizontal line.

The standard footnote marker is a small, raised number[1], which is sequentially numbered.

Footnotes produced with the `\footnote` command inside a **minipage** environment use the **mpfootnote** counter and are typeset at the bottom of the parbox produced by the **minipage**[2].

However, if you use the `\footnotemark` command in a `minipage` it will produce a footnote mark in the same style and sequence as the main text footnotes—i.e., stepping the `mpfootnote` counter and using the `\thefootnote` command for the representation. This behavior allows you to produce a footnote inside your `minipage` that is typeset in sequence with the main text footnotes at the bottom of the page: you place a `\footnotemark` inside the `minipage` and the corresponding `\footnotetext` after it. See below:

Footnotes in a minipage are numbered using lowercase letters.[a]
This text references a footnote at the bottom of the page.[3]

[a] Inside minipage

```
\begin{minipage}{5cm}
Footnotes in a minipage are numbered
using lowercase letters.\footnote{%
Inside minipage} \par This text
references a footnote at the bottom
of the page.\footnotemark
\end{minipage}
\footnotetext{At bottom of page}
```

The footnote numbering is incremented throughout the document for the article class, where it is reset to 1 for each new chapter in the report and book classes.

[1] See how the footnote is produced: " ... raised number `\footnote{See how the footnote is produced: ... }`.

[2] With nested minipages, the footnote comes after the next `\endminipage` command, which could be at the wrong place.

[3] At bottom of page.

XIII.1.1. Footnotes in tabular material

Footnotes appearing inside tabular material are not typeset by standard LaTeX. Only `tabularx` and `longtable` environments will treat footnotes correctly. But footnotes used in these tables won't appear just following the tables, but would appear at the bottom of the page just like the footnotes used in the text. But in `longtable` you can place the footnotes as table notes by placing the longtable in a minipage. See below:

Table XIII.1: PostScript type 1 fonts

Courier[a]	cour, courb, courbi, couri
Nimbus[b]	unmr, unmrs
URW Antiqua[b]	uaqrrc
URW Grotesk[b]	ugqp
Utopia[c]	putb, putbi, putr, putri

[a]Donated by IBM.
[b]Donated by URW GmbH.
[c]Donated by Adobe.

```
\begin{minipage}{.47\textwidth}
 \renewcommand{\thefootnote}{\thempfootnote}
  \begin{longtable}{ll}
   \caption{PostScript type 1 fonts}\\
    Courier\footnote{Donated by IBM.} & cour,courb,courbi,couri \\
    Nimbus\footnote{Donated by URW GmbH.} & unmr, unmrs \\
    URW Antiqua\footnotemark[\value{mpfootnote}] & uaqrrc\\
    URW Grotesk\footnotemark[\value{mpfootnote}] & ugqp\\
    Utopia\footnote{Donated by Adobe.} & putb, putbi, putr, putri
  \end{longtable}
 \end{minipage}
```

You can also put your `tabular` or `array` environment inside a `minipage` environment, since in that case footnotes are typeset just following that environment. Note the redefinition of `\thefootnote` that allows us to make use of the `\footnotemark` command inside the `minipage` environment. Without this redefinition `\footnotemark` would have generated a footnote mark in the style of the footnotes for the main page.

```
\begin{minipage}{.5\linewidth}
 \renewcommand{\thefootnote}{\thempfootnote}
  \begin{tabular}{ll}
   \multicolumn{2}{c}{\bfseries PostScript type 1 fonts} \\
   Courier\footnote{Donated by IBM.} & cour,courb,courbi,couri \\
   Charter\footnote{Donated by Bitstream.} & bchb,bchbi,bchr,bchri\\
   Nimbus\footnote{Donated by URW GmbH.} & unmr, unmrs \\
   URW Antiqua\footnotemark[\value{mpfootnote}] & uaqrrc\\
   URW Grotesk\footnotemark[\value{mpfootnote}] & ugqp\\
   Utopia\footnote{Donated by Adobe.} & putb, putbi, putr, putri
 \end{tabular}
\end{minipage}
```

PostScript type 1 fonts

Courier[a]	cour, courb, courbi, couri
Charter[b]	bchb, bchbi, bchr, bchri
Nimbus[c]	unmr, unmrs
URW Antiqua[c]	uaqrrc
URW Grotesk[c]	ugqp
Utopia[d]	putb, putbi, putr, putri

[a]Donated by IBM.
[b]Donated by Bitstream.
[c]Donated by URW GmbH.
[d]Donated by Adobe.

Of course this approach does not automatically limit the width of the footnotes to the width of the table, so a little iteration with the `minipage` width argument might be necessary.

Another way to typeset table notes is with the package `threeparttable` by Donald Arseneau. This package has the advantage that it indicates unambiguously that you are dealing with notes inside tables and, moreover, it gives you full control of the actual reference marks and offers the possibility of having a caption for our tabular material. In this sense, the `threeparttable` environment is similar to the nonfloating `table` environment.

```
\begin{threeparttable}
 \caption{\textbf{PostScript type 1 fonts}}
  \begin{tabular}{ll}
   Courier\tnote{a} & cour, courb, courbi, couri\\
   Charter\tnote{b} & bchb, bchbi, bchr, bchri \\
   Nimbus\tnote{c} & unmr, unmrs \\
   URW Antiqua\tnote{c} & uaqrrc\\
   URW Grotesk\tnote{c} & ugqp\\
   Utopia\tnote{d} & putb, putbi, putr, putri
  \end{tabular}
 \begin{tablenotes}
  \item[a] Donated by IBM.
  \item[b] Donated by Bitstream.
  \item[c] Donated by URW GmbH.
  \item[d] Donated by Adobe.
 \end{tablenotes}
\end{threeparttable}
```

Table 14.2: **PostScript type 1 fonts**

Courier[a]	cour, courb, courbi, couri
Charter[b]	bchb, bchbi, bchr, bchri
Nimbus[c]	unmr, unmrs
URW Antiqua[c]	uaqrrc
URW Grotesk[c]	ugqp
Utopia[d]	putb, putbi, putr, putri

[a] Donated by IBM.
[b] Donated by Bitstream.
[c] Donated by URW GmbH.
[d] Donated by Adobe.

XIII.1.2. Customizing footnotes

If the user wishes the footnote numbering to be reset to 1 for each `\section` command with the article class, this may be achieved by putting

```
\setcounter{footnote}{0}
```

before every section or using the following command at preamble[4]

```
\@addtoreset{footnote}{section}
```

The internal footnote counter has the name `footnote`. Each call to `\footnote` increments this counter by one and prints the new value in Arabic numbering as the footnote marker. A different style of marker can be implemented with the command

```
\renewcommand{\thefootnote}{number_style}{footnote}
```

where *number_style* is one of the counter print commands; `\arabic`, `\roman`, `\Roman`, `\alph`, or `\Alph`. However, for the counter `footnote`, there is an additional counter print command available, `\fnsymbol`, which prints the counter values 1–9 as one of nine symbols:

⋆ † ‡ § ¶ ‖ ⋆⋆ †† ‡‡

It is up to the user to see that the footnote counter is reset to zero sometime before the tenth `\footnote` call is made. If the user wants to add values above nine, then he has to edit the definition of `\fnsymbol`. See an example, which allows up to 12 footnotes without resetting the counter:

```
\makeatletter
\def\@fnsymbol#1{\ensuremath{\ifcase#1\or *\or \dagger\or \ddagger\or
   \mathsection\or \mathparagraph\or \|\or **\or \dagger\dagger
   \or \ddagger\ddagger\or \mathsection\mathsection
   \or \mathparagraph\mathparagraph \or \|\|\else\@ctrerr\fi}}
\renewcommand{\thefootnote}{\fnsymbol{footnote}}
\makeatother
```

An optional argument may be added to the `\footnote` command:

`\footnote[`*num*`]{`*footnote_text*`}`

where *num* is a positive integer that is used instead of the value of the footnote counter for the marker. In this case, the footnote counter is not incremented. For example**,

```
\renewcommand{\thefootnote}{\fnsymbol{footnote}}
  For example\footnote[7]{The 7$^{\rm th}$ symbol .... marker.},
\renewcommand{\thefootnote}{\arabic{footnote}}
```

where the last line is necessary to restore the footnote marker style to its standard form. Otherwise, all future footnotes would be marked with symbols and not with numbers.

XIII.1.3. Footnote style parameters

The appearance of the standard footnote can be changed by customizing the parameters listed below:

`\footnotesize` The font size used inside footnotes.

[4] This command will only work within `\makeatletter` and `\makeatother`.

** The 7th symbol appears as the footnote marker.